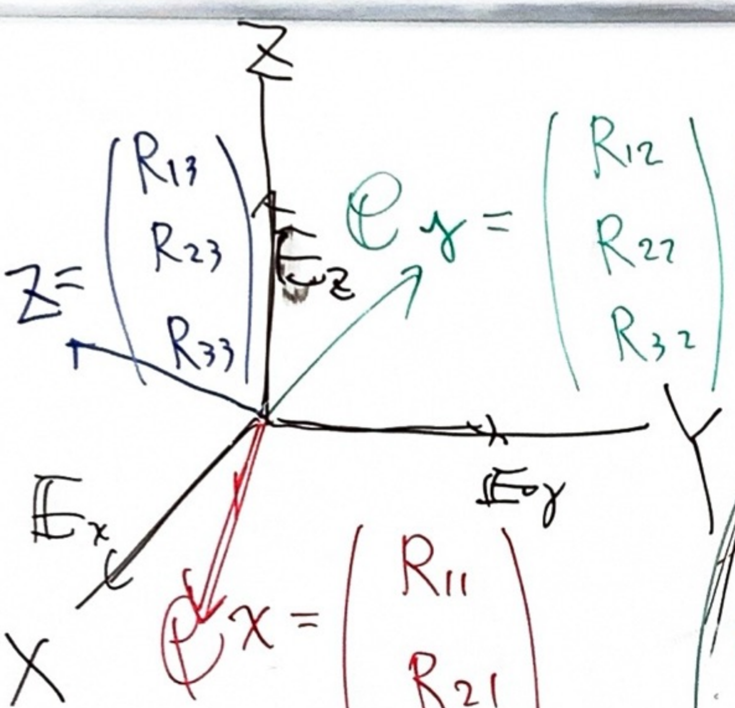

$$e_z = \begin{pmatrix} R_{13} \\ R_{23} \\ R_{33} \end{pmatrix} \qquad e_y = \begin{pmatrix} R_{12} \\ R_{22} \\ R_{32} \end{pmatrix} \qquad e_x = \begin{pmatrix} R_{11} \\ R_{21} \\ R_{31} \end{pmatrix}$$

基底変換行列

$$\overbrace{(e_x\ e_y\ e_z)}^{\text{基底(機体)}} = \overbrace{(E_x\ E_y\ E_z)}^{\text{基底(基準)}} \overbrace{\begin{pmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{pmatrix}}^{R}$$

$$e_x = \begin{pmatrix} R_{11} \\ R_{21} \\ R_{31} \end{pmatrix} = R_{11} E_x + R_{21} E_y + R_{31} E_z$$

$$e_y = R_{12} E_x + R_{22} E_y + R_{32} E_z$$

$$e_z = R_{13} E_x + R_{23} E_y + R_{33} E_z$$

$$v = \overbrace{(e_x\ e_y\ e_z)}^{\text{基底(機体)}} \begin{pmatrix} c_x \\ c_y \\ c_z \end{pmatrix} = (E_x\ E_y\ E_z) \begin{pmatrix} & & \\ & R & \\ & & \end{pmatrix} \begin{pmatrix} c_x \\ c_y \\ c_z \end{pmatrix}$$

$$= (E_x\ E_y\ E_z) \begin{pmatrix} d_x \\ d_y \\ d_z \end{pmatrix}$$

$$\begin{pmatrix} d_x \\ d_y \\ d_z \end{pmatrix} = \begin{pmatrix} R \end{pmatrix} \begin{pmatrix} c_x \\ c_y \\ c_z \end{pmatrix}$$